Enterprise Sales Intelligence

Complete Project Workflow

1. Project Introduction

Enterprise Sales Intelligence is a data analytics and business intelligence project developed using Microsoft AdventureWorks sales data, SQL Server, and Power BI.

The main objective is to transform raw sales data into meaningful business insights through data preparation, database management, data modeling, dashboard development, and automated data loading.

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2. Dataset Acquisition

Tool: Microsoft AdventureWorks Dataset

Activities performed:

1. Downloaded the AdventureWorks Sales dataset in Excel workbook format.
2. Opened the workbook and explored its available tables.
3. Identified seven tables required for sales analysis.

Seven source tables:

- Customer
- Date
- Product
- Reseller
- Sales
- Sales Order
- Sales Territory

Output: Original AdventureWorks Excel dataset.

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3. Conversion of Excel Tables into CSV Files

Tool: Microsoft Excel

Activities performed:

1. Opened the downloaded AdventureWorks workbook.
2. Selected each of the seven tables separately.
3. Exported each table into an individual CSV file.
4. Saved all seven CSV files in a dedicated folder.

CSV files created:

1. Customer data.csv
2. Date data.csv

3. Product data.csv
4. Reseller data.csv
5. Sales data.csv
6. Sales_order data.csv
7. Sales_territory data.csv

Output: Seven separate CSV files ready for database import.

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4. SQL Server Database Creation and Data Import

Tools: SQL Server 2025 and SQL Server Management Studio (SSMS)

Database Name: EnterpriseSalesDB

Activities performed:

1. Installed and configured SQL Server Database Engine.
2. Installed SQL Server Management Studio.
3. Created the EnterpriseSalesDB database.
4. Prepared a dedicated folder containing the seven CSV files.
5. Configured the SQL Server service account permissions to access the CSV folder.
6. Imported all seven CSV files into SQL Server.
7. Created corresponding database tables.
8. Verified the imported data using SQL queries.

Imported table record counts:

Table	Records
Customer data	18,485
Date data	1,461
Product data	397
Reseller data	702
Sales data	121,253
Sales_order data	121,253
Sales_territory data	11

Total imported records: 283,562

Output: Seven tables successfully imported into EnterpriseSalesDB.

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5. SQL Server to Power BI Data Connection

Tool: Power BI Desktop

Activities performed:

1. Opened Power BI Desktop.
2. Selected Get Data.
3. Chose SQL Server as the data source.
4. Connected to the SQL Server Database Engine.
5. Selected the EnterpriseSalesDB database.
6. Imported the required seven SQL Server tables into Power BI.
7. Verified the imported data.

Output: SQL Server data available inside Power BI for analysis.

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6. Data Transformation and Cleaning

Tool: Power Query Editor

After importing the SQL Server tables, data preparation was performed in Power BI.

Activities performed:

- Renamed columns for better readability.
- Corrected data types.
- Converted appropriate columns into date, text, and numeric formats.
- Replaced NA values in the Product Color column with Unknown.
- Applied required data cleaning and transformation steps.
- Prepared the tables for data modeling.

Output: Cleaned and structured data ready for analysis.

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7. Data Modeling

Tool: Power BI Model View

Activities performed:

1. Identified fact and dimension tables.
2. Established relationships between relevant tables.
3. Created a Star Schema-based data model.
4. Connected the Sales data with the relevant dimensions.
5. Reviewed relationships and data consistency.

Main tables used:

- Fact: Sales data
- Dimensions: Customer, Date, Product, Reseller, Sales Territory

Sales Order data was also imported and retained for the project.

Output: Structured relational data model for business intelligence reporting.

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8. DAX Measures and KPI Development

Created business measures to calculate important sales performance indicators.

Main KPIs:

- Total Sales
- Total Orders
- Total Profit
- Profit Margin

Dashboard KPI results recorded:

KPI	Value
Total Sales	109.81 Million
Total Orders	31K
Total Profit	12.55 Million
Profit Margin	Approximately 11.43%

These measures support business performance analysis.

- **Introduction**

DAX (Data Analysis Expressions) was used in Power BI to create calculated measures for sales performance analysis, profitability evaluation, time-based comparisons, and business intelligence reporting.

A combination of basic KPI measures and advanced time-intelligence calculations was implemented.

- **Basic DAX Measures**

1. Total Sales

Total Sales = SUM(Sales[Sales Amount])

Purpose: Calculates the overall sales revenue.

2. Total Profit

Total Profit =

SUM(Sales[Sales Amount]) - SUM(Sales[Total Product Cost])

Purpose: Calculates total profit by subtracting total product cost from sales amount.

3. Total Orders

Total Orders = DISTINCTCOUNT(SalesOrder[Sales Order])

Purpose: Counts the unique sales orders.

4. Profit Margin %

Profit Margin % =

DIVIDE([Total Profit], [Total Sales], 0)

Purpose: Calculates the percentage of profit generated from total sales.

- **Advanced DAX Measures**

5. Previous Year Sales

```
Previous Year Sales =
CALCULATE(
   [Total Sales],
   SAMEPERIODLASTYEAR('Date'[Date])
)
```

Purpose: Calculates sales for the corresponding period in the previous year.

6. YoY Growth %

```
YoY Growth % =
DIVIDE(
   [Total Sales] - [Previous Year Sales],
   [Previous Year Sales],
   0
)
```

Purpose: Measures the percentage increase or decrease in sales compared with the previous year.

7. YTD Sales (Year-to-Date Sales)

```
YTD Sales =
TOTALYTD(
   [Total Sales],
   'Date'[Date]
)
```

Purpose: Calculates cumulative sales from the beginning of the year up to the selected date.

8. Rolling 3-Month Average Sales

```
Rolling 3-Month Average Sales =
DIVIDE(
   CALCULATE(
      [Total Sales],
      DATESINPERIOD(
         'Date'[Date],
         MAX('Date'[Date]),
         -3,
         MONTH
```

)

),

3

)

Purpose: Calculates the average monthly sales over the rolling three-month period.

- **Data Validation and Supporting Measures**

9. Total Sales Rows

Total sales rows = COUNTROWS(Sales)

Purpose: Counts the total number of records in the Sales table.

10. Test Sales Rows

Test sales rows = COUNTROWS(Sales)

Purpose: Used as a supporting measure to validate the number of sales records.

- **Business Applications**

DAX Measure	Business Application
Total Sales	Revenue monitoring
Total Profit	Profitability analysis
Total Orders	Order volume analysis
Profit Margin %	Profit performance evaluation
Previous Year Sales	Historical comparison
YoY Growth %	Annual growth analysis
YTD Sales	Cumulative yearly performance
Rolling 3-Month Average Sales	Short-term sales trend analysis
Total sales rows	Sales record count
Test sales rows	Data validation

- **Conclusion**

DAX measures enhanced the analytical capabilities of the Enterprise Sales Intelligence dashboard. Basic KPI calculations provided an overview of business performance, while advanced time-intelligence measures enabled year-over-year comparisons, cumulative sales analysis, and rolling average calculations.

These calculations helped transform transactional sales data into meaningful business insights.

9. Power BI Dashboard Development

Tool: Power BI Desktop

Developed an interactive Enterprise Sales Intelligence dashboard.

Dashboard components:

- Total Sales KPI Card
- Total Orders KPI Card
- Total Profit KPI Card
- Profit Margin KPI Card
- Sales by Region
- Top 10 Products
- Top 10 Resellers
- Date and Region slicers
- Drill-through analysis page

Design and formatting:

- Applied a professional navy, white, and cyan color theme.
- Used KPI cards and charts for clear presentation.
- Applied consistent formatting.
- Added interactive slicers for filtering.
- Created a separate region-based drill-through page.

Output: Interactive Power BI sales analytics dashboard.

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10. SQL Server Data Movement Automation

Tools: SQL Server Management Studio and SQL Server Agent

To automate the loading of updated CSV data into the SQL Server database, an automation process was implemented.

Stored Procedure:

dbo.usp_Automate_EnterpriseSales_Load

SQL Server Agent Job:

EnterpriseSales_Automated_Data_Load

Activities performed:

1. Created a stored procedure for automated CSV data loading.
2. Prepared staging tables for the incoming data.
3. Used BULK INSERT to load CSV records into staging tables.

4. Included data validation and error handling.
5. Prepared the procedure to transfer the staged data into the main database tables.
6. Configured a SQL Server Agent Job to execute the stored procedure.
7. Configured a daily schedule at 3:00 AM.
8. Started the SQL Server Agent Job for execution.

Automation schedule: Daily at 3:00 AM.

Purpose: To automate the movement of the latest CSV data into SQL Server without manually importing each file.

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11. Power BI Data Refresh

After the automated SQL Server data loading process:

1. SQL Server Agent executes the scheduled automation job.
2. The stored procedure processes the CSV files.
3. The latest data is loaded into the SQL Server database.
4. Power BI Desktop is opened.
5. The user selects Home → Refresh.
6. Power BI retrieves the latest data from SQL Server.
7. The dashboard reflects the refreshed data.
8. The updated PBIX file is saved.

Important: SQL Server data loading is scheduled automatically. Power BI Desktop refresh is currently manual. Power BI Service publishing and scheduled dashboard refresh have not been configured.

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13. Technologies Used

Technology	**Purpose**
Microsoft Excel	Dataset exploration and CSV conversion
CSV	Data exchange format
SQL Server 2025	Database storage and management
SSMS	SQL queries, scripts and database operations
SQL Server Agent	Scheduled automation
SQL Stored Procedures	Automated data loading logic
BULK INSERT	CSV data loading
Power BI Desktop	Data analytics and dashboard development

Power Query	Data cleaning and transformation
DAX	KPI and business measure calculations
GitHub	Project source code and documentation storage

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14. Project Outcome

The project demonstrates an end-to-end business intelligence workflow, starting from raw AdventureWorks data and ending with an interactive sales analytics dashboard.

It combines SQL Server database management, automated CSV data loading, Power BI data modeling, DAX calculations, and interactive visual reporting.

The implemented SQL Server Agent automation reduces the need for repetitive manual CSV imports, while manual Power BI refresh retrieves the latest available database data.